UNIT - V

CORRELATION AND REGRESSION ANALYSIS

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INTRODUCTION

In this unit you will be able to learn the concept of correlation and regression. Also from this unit you will be able to learn the various methods of obtaining the correlation coefficients, rank correlation coefficient, regression equations etc. This unit explains the differences between the correlation and regression. It is easy to understand the techniques to be discussed in this unit by making use of calculation. Try out the example problems with the calculator.

5.1 MEANING OF CORRELATION

We are familiar that, the change in one factor, say, the amount of rainfall affects the change in the other factors, say, yield of rice. This means that there exist some kind

of relationship between the two factors. Thus correlation is the relationship between two factors.

In simple words, correlation means ***" the degree of relationship between two or more factors".*** An example of the relationship that exist between the price and demand.

5.2 TYPES OF CORRELATION

There are different types of correlation. They can be classified into the following categories.

a) Positive and Negative degree correlation

b) Linear and Non-linear correlation

First we will discuss positive and negative degree correlation.

5.2.1 Positive and Negative degree correlation

If the changes in the factors are in the same direction then the correlation is said to be ***"Positive degree correlation".*** Relationship between the amount of rainfall and yield of rice is an example of positive degree correlation. If the rainfall level increases then the yield of rice also increases and vice – versa.

If the changes in the factors are in the opposite direction then the correlation is said to be ***"Negative degree correlation".*** Relationship between the price and demand is an example of negative degree correlation. If the price level increases then the demand will decrease and vice – versa.

Now, we will discuss the linear and Non-linear correlation.

5.2.2 Linear and Non-linear Correlation

If the changes in the factors are in the constant ratio then the correlation is said to be ***"Linear correlation".***

For example,

Amount of rainfall (in mm)	40	60	80	100	120
Yield of rice(in tones)	100	150	200	250	300

From the above example, it can be observed that, amount of rainfall increases with 20mm at each level and yield of rice increases with 50 tonnes at each level.

If the changes in the factors are not in the constant ratio then the correlation is said to be ***"Non-linear correlation".***

For example,

Factor 1	***40***	***60***	***80***	***100***	***120***
Factor 2	80	120	130	140	205

From the above example, it can be observed that, the changes at various levels are different.

Self Assessment Question

State the different ***types of correlation*** with example in the space given below. Limit your answer in about 80 words.

_____

_____

_____

_____

5.3.1 Scatter Diagram

If the values of variables or factors, say, X and Y are plotted in the XY – plane, the diagram of the data obtained is called as ***scatter diagram.*** The greater the scatter of the plotted points on the diagram, the lesser is the relationship between the two variables or factors.

1. If all the points lie on a straight line falling from the lower left-hand corner to the upper right-hand corner, the correlation is said to be perfectively positive Fig. 5.1 (i.e., the correlation coefficient r = +1)

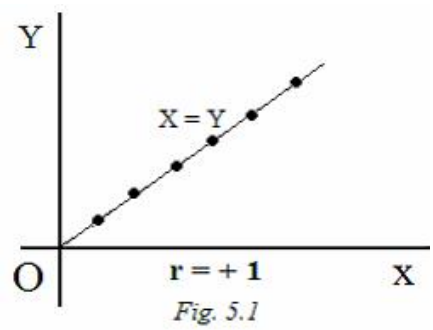

Fig. 5.1

2. *If all the points lie on a straight line falling from the upper left-hand corner to the upper right-hand corner, the correlation is said to be perfectively negative (i.e., the correlation coefficient r = - 1) Fig. 5.2*

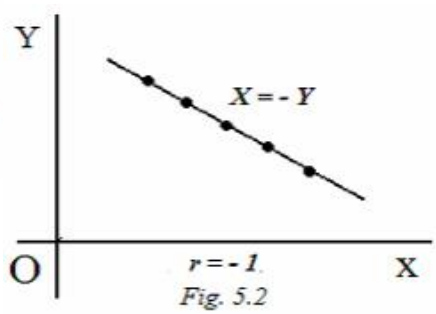

Fig. 5.2

3. *If all the points lie on a straight line fall in a narrow band and they show a rising tendency from the lower left-hand corner to the upper right-hand corner, there would be high degree of positive correlation. Fig.5.3*

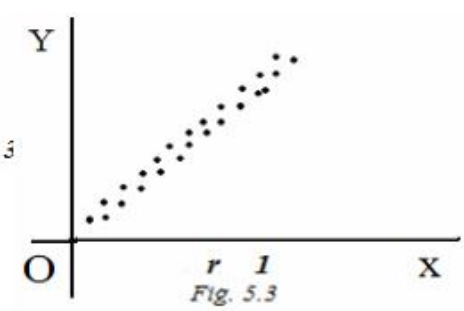

Fig. 5.3

4. *If all the points lie on a straight line fall in a narrow band and they show a declining tendency from the upper left-hand corner to the lower right-hand corner, there would be high degree of negative correlation Fig. 5.4*

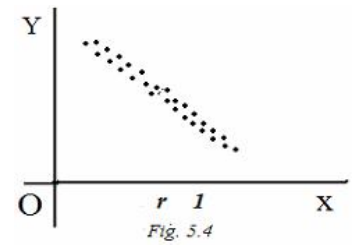

Fig. 5.4

5. *If all the points lie on a straight line fall in a widely band and they show a rising tendency from the lower left-hand corner to the upper right-hand corner, there would be low degree of positive correlation. Fig. 5.5*

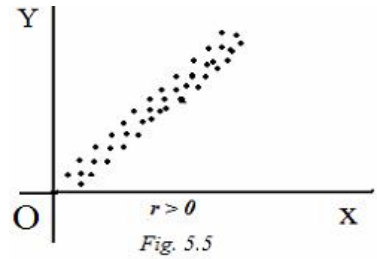

Fig. 5.5

6. *If all the points lie on a straight line fall in a widely band and they show a declining tendency from the upper left-hand corner to the lower right-hand corner, there would be low degree of negative correlation. Fig.5.6*

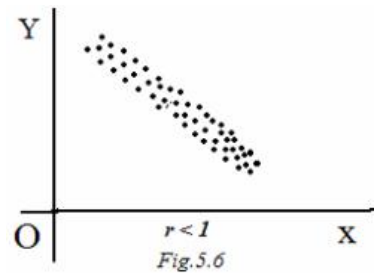

Fig.5.6

7. *If the plotted points lie on a straight line parallel to x-axis or in haphazard manner, it shows the absence of correlation between two factors. Fig. 5.7*

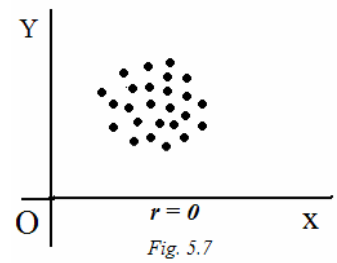

Fig. 5.7

5.3.2 KARL PEARSON'S COEFFICIENT OF CORRELATION

As a measure of degree of linear relationship between two variables, Karl Pearson developed a formula called correlation coefficient. The correlation coefficient between two variables usually denoted by r_{xy}, is a measure of relationship between them, is defined as,

$$r_{xy} = \frac{Cov(x, y)}{s_x s_y}$$

$$= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{\sqrt{\sum (X - \overline{X})^2}\sqrt{\sum (Y - \overline{Y})^2}}$$

$$= \frac{\sum xy}{\sqrt{\sum x^2}\sqrt{\sum y^2}} \qquad \text{where } x = X - \overline{X};\ y = Y - \overline{Y}$$

Working Procedure

Step 1: Denote one series by X and the other by Y.

Step 2 : Calculate $\bar{X}$ and $\bar{Y}$ of the X and Y series respectively, using the formula, $\bar{X} = \frac{\sum X}{n}$; and $\bar{Y} = \frac{\sum Y}{n}$

Step 3 : Take the deviations of the observations in X-series from $\bar{X}$ and write it under column headed by $x = X - \bar{X}$. Take the deviations of the observations in Y series from $\bar{Y}$ and write it under the column headed by $y = Y - \bar{Y}$.

Step 4 : Multiply the respective deviations and write it under the column headed by xy

Step 5 : Square the deviations obtained in step 4 for X & Y series and write it under the column headed by x^2 and y^2 .

Step 6 : Apply the following formula to calculate the correlation coefficient (r)

$$r_{xy} = \frac{\text{å } xy}{\sqrt{\text{å } x^2}\sqrt{\text{å } y^2}}$$

Example 5.1 Find the coefficient of correlation between height of brothers and sisters from the following data.

Heights of Brothers (in cm)	65	66	67	68	69	70	71
Heights of Sisters (in cm)	67	68	66	69	72	72	69

Solution : Let the heights of Brothers be denoted by X and that of Sisters by Y. Let us prepare the following table

X	$(x = X - \bar{X})$	Y	$(y = Y - \bar{Y})$	xy	x^2	y^2
65	-3	67	-2	6	9	4
66	-2	68	-1	2	4	1
67	-1	66	-3	3	1	9
68	0	69	0	0	0	0
69	1	72	+3	+3	1	9
70	2	72	+3	+6	4	9
71	3	69	0	0	9	0
476	**-**	**483**	**-**	**20**	**28**	**32**

From the above table,

n = 7, $\sum X$ = 476; $\sum Y$ = 483; $\sum xy$ = 20, $\sum x^2$ = 28, Σy^2 = 32,

$$\overline{X} = \frac{\sum X}{n} = \frac{476}{7} = 68$$

$$\overline{Y} = \frac{\Sigma Y}{n} = \frac{483}{7} = 69$$

Karl Pearson's Coefficient of Correlation is now calculated as follows:

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2}\sqrt{\Sigma y^2}}$$

$$= \frac{20}{\sqrt{28}\sqrt{32}}$$

$$= \frac{20}{(5.2915)(5.6569)} = \frac{20}{29.9335}$$

$$r = 0.6681$$

Self-Assessment Question

Calculate the correlation coefficient between the height of sisters and height of brothers from the given data :

Height of Sisters (in cm)	64	65	66	67	68	69	70
Height of Brothers (in cm)	66	67	65	68	70	68	72

[Hint : $\overline{X} = 67, \overline{Y} = 68,\ \Sigma x^2 = 28, \Sigma y^2 = 34,\ \Sigma xy = 25, r = 0.81$]

Short Cut Method

The above direct method for calculating '**r**' is not convenient when (i) the terms of the series X and Y are larger and the calculation of X and Y becomes difficult (or) (ii) the mean of X or Y are not integers. In these cases we apply the following formula of assumed mean.

$$r_{xy} = \frac{n\Sigma dxdy - (\Sigma dx)(\Sigma dy)}{\sqrt{n\Sigma dx^2 - (\Sigma dx)^2}\sqrt{n\Sigma dy^2 - (\Sigma dy)^2}}$$

where, d x = X – A, A is the assumed mean of X - series.

d y = Y - B, B is the assumed mean of Y - series.

n - is number of observation of X and Y.

Working Procedure

Step 1: Denote one series by X and the other by Y.

Step 2 : Take any term 'A' as assumed mean of X series and 'B' as assumed mean of Y series. (preferably the middle one)

Step 3 : Take the deviations of the observations in X-series from A and write it under coloumn headed by dx = X – A. Take the deviations of the observations in Y series from B and write it under the coloumn headed by dy = Y - B.

Step 4 : Multiply the respective deviations and write it under the coloumn headed by dx dy

Step 5 : Square the deviations obtained in step 4 for X & Y series and write it under the coloumn headed by dx^2 and dy^2.

Step 6 : Apply the following formula to calculate the correlation coefficient(r)

$$r_{xy} = \frac{n\Sigma dxdy - (\Sigma dx)(\Sigma dy)}{\sqrt{n\Sigma dx^2 - (\Sigma dx)^2}\sqrt{n\Sigma dy^2 - (\Sigma dy)^2}}$$

Example 5.2: Calculate the coefficient of correlation for the following pairs of values of X and Y

X	**17**	**19**	**21**	**26**	**20**	**28**	**26**	**29**
Y	23	27	25	26	27	25	30	33

Solution :

Let the assumed means for X and Y be 23 and 27 respectively, so that, $d_x = X - 23$, $d_y = Y - 27$, we have the following table.

X	Y	$d_x = X - 23$	$d_y = Y - 27$	$d_x d_y$	d_x^2	d_y^2
17	23	-6	-4	24	36	16
19	27	-4	0	0	16	0
21	25	-2	-2	4	4	4
26	26	3	-1	-3	9	1
20	27	-3	0	0	9	0
28	25	5	-2	-10	25	4
26	30	3	3	9	9	9
29	33	6	6	36	36	36
186	**216**	**2**	**0**	**60**	**144**	**70**

Note that, here $\overline{X} = \frac{\Sigma X}{n} = \frac{186}{8} = 23.25$, which is not an integer, we use short-cut method.

Here n = 8; $\Sigma dx = 2; \Sigma dy = 0; \Sigma dxdy = 60; \Sigma dx^2 = 144; \Sigma dy^2 = 70$

$$r_{xy} = \frac{n\Sigma dxdy - (\Sigma dx)(\Sigma dy)}{\sqrt{n\Sigma dx^2 - (\Sigma dx)^2}\sqrt{n\Sigma dy^2 - (\Sigma dy)^2}}$$

$$r_{xy} = \frac{8(60)-(2)(0)}{\sqrt{8(144)-(2)^2}\sqrt{8(70)-(0)^2}} = \frac{480}{\sqrt{1148}\sqrt{560}} = \frac{480}{(33.8821)(23.6643)}$$

$$= \frac{480}{801.7962}$$

$$r_{xy} = 0.5987$$

Self-assessment Question

Compute the coefficient of correlation for the following data

X	**10**	**25**	**13**	**25**	**22**	**11**	**12**	**25**	**21**	**20**
Y	12	22	16	15	18	18	17	23	24	17

[Ans: r_{xy} = 0.53]

5.3.3 Spearman's Rank correlation coefficient

The coefficient of rank correlation is based on the various values of the variates and is denoted by ρ. It is applied in the problems in which data cannot be measured quantitatively but qualitative assessment is possible such as beauty, honesty etc. In this case the best individual is given rank number 1, next rank 2 and so on.

The Coefficient of rank correlation is given by the formula.

$$\rho = 1 - \frac{6\Sigma D^2}{n^3 - n}$$

Where D - is the difference of corresponding ranks and n - is the number of pairs of observations.

TYPE I : RANKS ARE GIVEN DIRECTLY

Working Procedure

Step 1 : Denote rank of X series by R_1 and rank of Y series by R_2.

Step 2 : Calculate the difference of R_1 and R_2 and write it under the column headed by D.

Step 3 : Square the difference D and write it under the column headed by D^2.

Step 4 : Apply the formula : $\rho = 1 - \frac{6\sum D^2}{n^3 - n}$

This method is described with the following example :

Example 5.3 : Two judges in a beauty contest rank the 12 entries as follows:

Judge X	1	2	3	4	5	6	7	8	9	10	11	12
Judge Y	12	9	6	10	3	5	4	7	8	2	11	1

Calculate the rank correlation coefficient between the two judges X and Y.

Solution :

Judge X (R_1)	***Judge Y (R_2)***	***$D = R_1 - R_2$***	***D^2***
1	12	-11	121
2	9	-7	49
3	6	-3	9
4	10	-6	36
5	3	2	4
6	5	1	1
7	4	3	9
8	7	1	1
9	8	1	1
10	2	8	64
11	11	0	0
12	1	11	121
	Total		**416**

Here, n = 12 ; $\Sigma D^2 = 416$

Now,

$$\rho = 1 - \frac{6\sum D^2}{n^3 - n}$$

$$= 1 - \frac{6(416)}{12^3 - 12} = 1 - \frac{2496}{1728 - 12}$$

$$= 1 - \frac{2496}{1716} = 1 - 1.4545$$

$$\rho = -0.4545$$

Example 5.4 : Ten competitors in a beauty contest were ranked by three judges in the following order:

Judge 1	1	6	5	10	3	2	4	9	7	8
Judge 2	3	5	8	4	7	10	2	1	6	9
Judge 3	6	4	9	8	1	2	3	10	5	7

Use the rank correlation coefficient to determine which pair of judges has the nearest approach to common taste in beauty.

Solution

Let R_1, R_2, R_3 respectively be the ranks given by first, second and third judge.

Let ρ_{ij} be the rank correlation coefficient between the ranks given by i^{th} and j^{th} judges, $i = 1,2,3$; $j = 1,2,3$.

Let $D_{ij} = Ri - Rj$, be the difference of ranks of an individual given by i^{th} and j^{th} judge.

Judge 1 R_1	***Judge2*** R_2	***Judge3*** R_3	$D_{12} = R_1 - R_2$	D_{12}^2	$D_{23} = R_2 - R_3$	D_{23}^2	$D_{13} = R_1 - R_3$	D_{13}^2
1	3	6	-2	4	-3	9	-5	25
6	5	4	1	1	1	1	2	4
5	8	9	-3	9	-1	1	4	16
10	4	8	6	36	-4	16	2	4
3	7	1	-4	16	6	36	2	4
2	10	2	-8	64	8	64	0	0
4	2	3	2	4	-1	1	1	1
9	1	10	8	64	-9	81	-1	1
7	6	5	-1	1	1	1	2	4
8	9	7	-1	1	2	4	1	1
		Total		**200**		**214**		**60**

Here, n = 10; $\sum D_{12}^2 = 200; \sum D_{23}^2 = 214; \sum D_{13}^2 = 60$

First & Second Judges

$$\rho_{12} = 1 - \frac{6\sum D_{12}^2}{n^3 - n} = 1 - \frac{6(200)}{10^3 - 10} = 1 - \frac{1200}{990}$$

$$\rho_{12} = 1 - 1.2121 = -0.2121$$

Second & Third Judges

$$\rho_{23} = 1 - \frac{6\sum D_{23}^2}{n^3 - n} = 1 - \frac{6-(214)}{10^3 - 10} = 1 - \frac{1284}{990} = 1 - 1.2969 = -0.2969$$

First & Third Judges

$$\rho_{13} = 1 - \frac{6\sum D_{13}^2}{n^3 - n}$$

$$\rho_{13} = 1 - \frac{6(60)}{10^3 - 10} = 1 - \frac{360}{990} = 1 - 0.3636$$

$$\rho_{13} = 0.6364$$

Since ρ_{13} is maximum, thus the pair of first and third judges has the nearest approach to common taste in beauty.

Self-assessment Question

1. Two judges in a musical contest rank the 10 entries as follows:

Judge X	3	5	8	4	7	10	2	1	6	9
Judge Y	6	4	9	8	1	2	3	10	5	7

[Hint: n = 10; $\sum D^2 = 149; \rho = 0.8495$]

2. Ten competitors in a beauty contest were ranked by three judges in the following order:

1st Judge	1	5	4	8	9	6	10	7	3	2
2nd Judge	4	8	7	6	5	9	10	3	2	1
3rd Judge	6	7	8	1	5	10	9	2	3	4

Use spearman's coefficient of rank correlation to determine which pair of judges has the nearest approach to common taste in beauty:

[Hint: $n = 10; \sum D_{12}^2 = 74; \sum D_{23}^2 = 44; \sum D_{13}^2 = 156;$

$\rho_{12} = 0.5515; \rho_{23} = 0.7333; \rho_{13} = 0.0545$]

TYPE II : RANKS ARE NOT GIVEN – NON – REPEATED RANKS

In this case we are given only the data. We assign the ranks to both the series of X and Y by giving the ranks in ascending order for both series (or descending order).

Working Rule

Step 1: Assign ranks to each items of both series in ascending or descending order.

Step 2: Calculate the difference of ranks and write it under the column headed by D.

Step 3: Square the difference D and write it under the heading D^2.

Step 4: Apply the formula, $\rho = 1 - \frac{6\sum D^2}{n^3 - n}$

This method is explained with the help of the following example.

Example 5.5

For the following data calculate the coefficient of rank correlation.

Series X	80	91	99	71	61	81	70	59
Series Y	123	135	154	110	105	134	121	106

Solution

Series X	***Series Y***	***Rank X (R_1)***	***Rank Y (R_2)***	***D***	***D^2***
80	123	5	5	0	0
91	135	7	7	0	0
99	154	8	8	0	0
71	110	4	3	1	1
61	105	2	1	1	1
81	134	6	6	0	0
70	121	3	4	- 1	1
59	106	1	2	- 1	1
Total					**4**

Here, n = 8 ; $\sum D^2 = 4$

Now, $$\rho = 1 - \frac{6\sum D^2}{n^3 - n}$$

$$= 1 - \frac{6(4)}{8^3 - 8}$$

$$= 1 - \frac{24}{504}$$

$$= 1 - 0.0476$$

$$\rho = 0.09524$$

Self-assessment Question

Calculate the rank correlation coefficient for the following data of two series.

Series X	92	89	87	86	83	77	71	63	53	50
Series Y	86	83	91	77	68	85	52	82	37	57

[Hint: n = 10; $\sum D^2 = 44$; $\rho = 0.733$]

TYPE III : RANKS ARE NOT GIVEN – REPEATED RANKS

If two or more individuals are placed together in any classification with respect to an attribute, there are more than one item with the same rank in either or both the series,

then the problem is solved by assigning average rank to each of their individuals who are put in a tie.

For example, suppose an item is repeated at rank 5, (*i.e.*, the 5th and 6th items are having same values), then the common rank assigned to 5th and 6th is (5+6)/2 = 5.5. The next rank assigned thrice, then the common rank assigned to the value is sum of the ranks by divided by 3. In order to find the rank correlation coefficient the adjustment factor is added to the formula, which is given by

Adjustment Factor (A.F.) = $\frac{1}{12}(m^3 - m)$

Where '*m*' is the number of times an item is repeated. This Adjustments Factor is to be added for each repeated value in both the series.

The modified formula for the rank correlation coefficient is given by,

$$\rho = 1 - \frac{6\left[\sum D^2 + \frac{1}{12}\sum(m^3 - m)\right]}{n^3 - n}$$

This method is explained with the following example,

Example 5.6

From the following data related to the series X and Y, calculate the coefficient of rank correlation.

Series X	48	33	40	9	16	16	65	24	16	57
Series Y	13	13	24	6	15	4	20	9	6	19

Solution

Series X	***Series Y***	***Rank X(R_1)***	***Rank Y(R_2)***	***D = $R_1 - R_2$***	***D^2***
48	13	8	5.5	2.5	6.25
33	13	6	5.5	0.5	0.25
40	24	7	10	-3	9.00
9	6	1	2.5	-1.5	2.25
16	15	3	7	-4	16.00
16	4	3	1	2	4.00
65	20	10	9	1	1.00
24	9	5	4	1	1.00
16	6	3	2.5	0.5	0.25
57	19	9	8	1	1.00
Total					**41.00**

Here n = 10, $\sum D^2 = 41$.

[*Remark* : In the X series, we see that the value 16 is repeated thrice, the common rank is given to the X value is 3, which is the average of 2,3 & 4.
i.e., (2+3+4)/3 = 3].

Now, Adjustment Factor

For X series, $AF_1 = \frac{1}{12}(3^3 - 3) = 2$

For Y series, $\left\{ \begin{array}{l} AF_2 = \frac{1}{12}(2^3 - 2) = 0.5 \\ AF_3 = \frac{1}{12}(2^3 - 2) = 0.5 \end{array} \right\}$

\ The coefficient of rank correlation is,

$$\rho = 1 - \frac{6\left[\sum D^2 + \frac{1}{12}\sum(m^3 - m)\right]}{n^3 - n}$$

$$= 1 - \frac{6\left[41 + 2 + 0.5 + 0.5\right]}{10^3 - 10}$$

$$= 1 - \frac{6[44]}{990}$$

$$= 1 - \frac{264}{990}$$

$$\rho = 1 - 0.2667$$

$$\rho = 0.7333$$

Self-assessment Question

Obtain the rank correlation coefficient for the following data.

Series X	68	64	75	50	64	80	75	40	55	64
Series Y	62	58	68	45	81	60	68	48	50	70

[Hint: n = 10; $\sum D^2 = 72; \rho = 0.545$]

5.4 Properties of Correlation Coefficient

- The value of 'r' does not depend on which of the two variables under study is labeled X and which is labeled Y.

- The correlation coefficient lies between –1 and +1. i.e., $-1 \le r \le +1$.
- The correlation coefficient is independent of change of origin and scale.
- r = +1, if all (X_i, Y_j) pairs lie on a straight line with positive slope and

  r = -1, if all (X_i, Y_j) pairs lie on a straight line with negative slope.

5.5 REGRESSION ANALYSIS

Managers often make decisions by studying the relationship between variables, and process improvements can often be made by understanding how changes in one or more variables affect the process output. Regression Analysis is a statistical technique in which we observe data to relate a variable of interest, which is called the ***dependent*** (or ***response***) variable, to one or more ***independent*** (or ***predictor***) variables. The objective is to build a regression model, or prediction equation, that can be used to describe, predict and control the dependent variable on the basis of the independent variables. For example, a company might wish to improve its marketing process. After collecting data concerning the demand for a product, the product's price, and the advertising expenditures made to promote the product, the company might use regression analysis to develop an equation to predict demand on the basis of price and advertising expenditure. Predictions of demand for various price-advertising expenditure combinations can then be used to evaluate potential changes in the company's marketing strategies.

In the words of M.M. Blair,

Regression analysis is a ***"mathematical measure of average relationship between two or more variables in terms of the original units of the data".***

5.5 Types of Regression Lines

A line of regression is the line, which gives the best estimate of one variable X, for any given value of the other variable. We have two types of regression lines, namely,

- ***Regression line of X on Y***
- ***Regression line of Y on X.***

First we will give the regression line of X on Y.

It is the line, which gives the best estimate for the values of X for a specified value of Y.

It is given by,

$$X - \bar{X} = b_{xy} (Y - \bar{Y})$$

Where b_{xy} is the regression coefficient of X on Y, which can be calculated using any of the formula under the natures of the data.

$$b_{xy} = \frac{\sum xy}{\sum y^2} \quad \text{where } x = X - \bar{X} \ \& \ y = Y - \bar{Y}$$

(or)

$$b_{xy} = \frac{n\sum d_x d_y - (\sum d_x)(\sum d_y)}{n\sum d_y^2 - (\sum d_y)^2}$$

where, $d_x = X - A;\ d_y = Y - B$; and A, B are assumed mean.

(or)

$$b_{xy} = r\frac{\sigma_x}{\sigma_y}$$

where 'r' is the correlation coefficient, σ_x and σ_y are the standard deviations for X and Y series.

Now we give the regression line of Y on X.

It is the line, which gives the best estimate for the values of Y for a specified value of X. It is given by.

$$Y - \bar{Y} = b_{yx}\ (X - \bar{X})$$

Where b_{yx} is the regression line of Y on X, which can be calculated using any one of the following formula depending upon the nature of the data.

$$b_{yx} = \frac{\sum xy}{\sum x^2}\ ;\ \text{where, } x = X - \bar{X}\ \&\ y = Y = \bar{Y}$$

(or)

$$b_{yx} = \frac{n\sum d_x d_y - (\sum d_x)(\sum d_y)}{n\sum d_x^2 - (\sum d_x)^2}$$ where, $d_x = X - A;\ d_y = Y - B$ and A, B are assumed mean.

(or)

$b_{yx} = r\dfrac{\sigma y}{\sigma x}$ where 'r' is the correlation coefficient σ_x, σ_y are the standard deviations of X and Y series.

5.7 CONSTRUCTION OF REGRESSION EQUATION

Example 5.7

The height of a sample of 10 fathers and their eldest sons are given below (to the nearest *cm*.).

Height of Father (X)	170	167	162	163	167	166	169	171	166	169
Height of Son (Y)	166	167	164	166	166	164	168	170	163	166

(i) Obtain the two regression equations

(ii) Estimate the likely height of Father when the height of Son is 190 *cm*.

(iii) Estimate the likely height of Son when the height of Father is 160 *cm*.

Solution

Height of Father (X)	**Height of Son (Y)**	$x = X - \bar{X}$ **(** $\bar{X}$ **= 167)**	$y = Y - \bar{Y}$ **(** $\bar{Y}$ **= 166)**	xy	x^2	y^2
170	166	+3	0	0	9	0
167	167	0	+1	0	0	1
162	164	-5	-2	10	25	4
163	166	-4	0	0	16	0
167	166	0	0	0	0	0
166	164	-1	-2	2	1	4
169	168	2	2	4	4	4
171	170	4	4	16	16	16
166	163	-1	-3	3	1	9
169	166	2	0	0	4	0
1670	**1660**	**0**	**0**	**35**	**76**	**38**

Here, n = 10, $\sum X$ = 1670, $\sum Y$ = 1660

$$\bar{X} = \frac{\sum X}{n} = \frac{1670}{10} = 167$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{1660}{10} = 166$$

From the table, $\sum xy$ = 35; $\sum x^2$ = 76; $\sum y^2$ = 38

$$b_{xy} = \frac{\sum xy}{\sum y^2} = \frac{35}{38} = 0.9211$$

$$b_{yx} = \frac{\sum xy}{\sum x^2} = \frac{35}{76} = 0.4605$$

(i) Regression line of X on Y

$$X - \bar{X} = b_{xy}\,(Y - \bar{Y})$$

$$X - 167 = 0.9211\,(Y - 166)$$

$$X - 167 = 0.9211\,Y - 152.9028$$
$$X = 0.9211Y - 152.9026 + 167$$
$$X = 0.9211Y + 14.934$$

Regression line of Y on X

$$Y - \bar{Y} = b_{yx}(X - \bar{X})$$
$$Y - 166 = 0.4605\,(X - 167)$$
$$Y - 166 = 0.4605\,X - 76.9035$$
$$Y = 0.4605\,X - 76.9035 + 166$$
$$Y = 0.4605X + 89.0965$$

(ii) Given, Height of Son (Y) = 190 *cm*. To estimate the height of Father (X), we use X on Y equation.

$$X = 0.9211\,Y + 14.0934$$
$$X = 0.9211(190) + 14.0934$$
$$X = 175.009 + 14.0934$$
$$X = 189.1024\,cm$$

(iii) Given, Height of Father (X) = 160 *cm*. To estimate the height of son, we use Y on X equation.

$$Y = 0.4605\,X + 89.0965$$
$$Y = 0.4605\,(160) + 89.0965$$
$$Y = 73.68 + 89.0965$$
$$Y = 162.78\,cm.$$

Self Assessment Question

From the following data, obtain the two regression equations.

Sales	91	97	108	121	67	124	51	73	111	57
Purchase	71	75	69	97	70	91	99	61	80	47

Also estimate the sales when the purchase is 90.

[Hint: n = 10; $\bar{X}$ = 90; $\bar{Y}$ = 76; $\sum xy$ = 3900; $\sum x^2$ = 6360; $\sum y^2$ = 2388;

b_{xy} = 1.36; b_{yx} = 0.6132 ; Line of X on Y : X = 1.36 Y – 5.2 ;

Line of Y on X : Y = 0.6132 X + 14.812;

Estimated sales, when the purchase is 90 = 117.2]

Example 5.8

Find the two lines of regression from the following data.

Price at Mumbai (in Rs.)	36	42	55	61	76	26
Price at Chennai (in Rs.)	15	36	24	26	15	14

Also estimate the likely price at Mumbai when the price at Chennai is Rs.60/-

Solution

Price at Mumbai (X)	**Price at Chennai (Y)**	dx **= X – A (A = 55)**	dy **=Y–B (B = 26)**	$dxdy$	dx^2	dy^2
36	15	-19	-11	209	361	121
42	36	-13	10	-130	169	100
55	24	0	2	0	0	4
61	26	6	0	0	36	0
76	15	21	-9	-189	441	81
26	14	-29	-12	348	841	144
296	**130**	**-34**	**-44**	**238**	**1848**	**450**

Here, n = 10; $\sum X$ = 296; $\sum Y$ = 130; $\sum d_x$ = -34; $\sum d_y$ = -44; $\sum d_x d_y$ = 238; $\sum d_x^2$ = 1848; $\sum d_y^2$ = 450.

$$\bar{X} = \frac{\sum X}{n} = \frac{296}{6} = 49.33$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{130}{6} = 21.67$$

$$b_{xy} = \frac{n\sum d_x d_y - (\sum d_x)(\sum d_y)}{n\sum d_y^2 - (\sum d_y)^2}$$

$$= \frac{6(238)-(-34)(-44)}{6(450)-(-44)^2} = \frac{1428-1496}{2700-1936} = \frac{-68}{764}$$

$$b_{xy} = -0.089$$

$$b_{xy} = \frac{n\sum dxdy - (\sum dx)(\sum dy)}{n\sum dx^2 - (\sum dx)^2}$$

$$= \frac{6(238)-(-34)(-44)}{6(1848)-(-34)^2} = \frac{-68}{9888-1150} = \frac{-68}{8732}$$

$$b_{xy} = 0.0078$$

Regression Line of X on Y

$$X - \overline{X} = b_{xy}\ (Y - \overline{Y})$$

X – 49.33 = - 0.089 (Y- 21.67)

X - 49.33 = - 0.089Y + 1.9286

X = - 0.089Y + 1.9286 + 49.33

X = - 0.089Y + 51.2586

Regression line of Y on X

$$Y - \overline{Y} = b_{yx}\ (X - \overline{X})$$

Y – 21.67 = - 0.0078 (X – 49.33)

Y - 21.67 = - 0.0078X + 0.3848

Y = - 0.0078X + 0.3848 + 21.67

Y = - 0.0078 X + 22.0548

To find the Estimate likely price at Mumbai, we use the line of X on Y

X = - 0.089Y + 51.2586

X = - 0.089(60) + 51.2586 = 45.92

Hence the price at Mumbai is Rs. 45.92

Self Assessment Question

Find the two lines of regression from the following data

Age of Husband	23	22	28	26	35	20	22	40	20	18
Age of Wife	18	15	20	17	22	14	16	21	15	14

Hence estimate, the age of husband when the age of wife is 19.

[Hint : n = 10; $\overline{X}$ = 25.6 $\overline{Y}$ = 17.2 ; b_{xy} = 2.23; b_{yx} = 0.385

X = 2.23Y– 12.76; Y = 0.385 X + 7.3Y ; Age of Husband (X) = 29.61.]

Example 5.9

Find out the likely production corresponding to a rainfall of 40cms from the following data:

	Rainfall (in cms.)	***Output (in quintals)***
Average	30	50
Standard deviation	5	10

Coefficient of correlation, r = 0.8.

Solution

Let X and Y denotes the rainfall and output respectively.

Given : $\overline{X} = 30;\ \overline{Y} = 50;\ \sigma_x = 5;\ \sigma_y = 10;\ r = 0.8$

Regression line of Y on X

$$Y - \overline{Y} = b_{yx}\ (X - \overline{X})$$

$$b_{yx} = r\left(\frac{\sigma_y}{\sigma_x}\right) = 0.8\left(\frac{10}{5}\right) = 1.6$$

Y- 50 = 1.6 (X – 30)

Y- 50 = 1.6X – 48

Y = 1.6X – 48 + 50

Y = 1.6X + 2

When rainfall X = 40 *cm*,

Y = 1.6 (40) + 2

Y = 66 Quintals.

Self Assessment Question

Estimate the most likely yield of paddy when the annual rainfall is 22 cms other factors being assumes to remain the same.

	Yield per hectare (in kgs.)	***Annual rainfall (in cms.)***
Mean	973.5	18.3
Standard deviation	38.4	2.0

Coefficient of Correlation = 0.58.

[Hint: Regression line of Y and X , Y = 11.136X + 769.71;

For X = 22; Yield(Y) = 1014.7 *kgs*.]

5.8 Properties of regression coefficients

1. There two regression lines, namely, X on Y and Y on X and they always intersect at the mean $(\overline{X}, \overline{Y})$
2. If one regression coefficient is greater than unity, then the other one has to be less than unity.
3. Geometric mean between the regression coefficients is correlation coefficient. (i.e., $r = \pm\sqrt{b_{xy}\ b_{yx}}$)
4. Although regression equations are usually different, they become identical if $r = +1$.
5. If $r = 0$ then the regression lines are perpendicular to each other.

5.9 Difference between correlation and regression

Correlation	*Regression*
1. It is the degree of relationship between two or more variables or factors.	1. It is the average relationship between two or more variables or factors.
2. It is symmetric in x and y i.e., $r_{xy} = r_{yx}$	2. The regression coefficients are not symmetric.
3. The correlation coefficient does not reflect upon the nature of variable. (independent or dependent variable)	3. Regression coefficients reflect on the nature of variable i.e., which is dependent and which is independent.
4. It does not imply cause and effect relationship; between the variables under study.	4. It indicates the cause and effect relationship between the variables. The variable corresponding to cause is taken as independent variable, whereas corresponding to effect is taken as dependent variable.
5. It is a relative measure and is independent of the units of measurement	5. Regression coefficients are absolute measure of finding out the relationship between two or more variables.
6. It indicates the degree of association.	6. It is used to forecast the nature of dependent variable when the value ofindependent variable is known.

5.10 Applications of Regression

- The cause and effect relations are indicated from the study of regression analysis.
- It establishes the rate of change in one variable in terms of the changes in another variable
- It is useful in economic analysis as regression equation can determine an increase in the cost of living index for a particular increase in general price level.
- It helps in prediction and thus it can estimate the values of unknown quantities.
- It helps in determining the coefficient of correlation.
- It enable us to study the nature of relationship between the variables.
- It can be useful to all natural, social and physical sciences, where the data are in functional relationship.

Chapter Summary

This chapter has discussed simple correlation coefficient, correlation coefficient and simple linear regression analysis, which relates a dependent variable to a single independent variable. We began by considering the simple linear regression model, which employs two parameters; the slope and y intercept. We next discussed how to compute the least square point estimates of these parameters and how to construct the regression equations by using various methods. We learned that the differences between correlation and regression and applications of regression analysis.